

15 years. If you look at the ARPANET logical map (see Figure 2), you can see how simple (!) the network was. Back then, nobody thought that we might run out of IPv4 unique values and have to go for alternate options like IPv6. This indicates that the layers can change in future to improve the performance of the network.

Though I mentioned TCP and UDP while discussing Layer 4, I think it is necessary to discuss them a bit more. The TCP/IP suit is one of the core protocols we use in the Internet. The IP is largely responsible for the delivery of the data by taking it all the way through the network. But TCP is concerned with issues only at the source and at the destination. A typical example could be a Web server and a browser. Please refer to Figure 3 for information concerning its header.

UDP can be employed to send messages to other hosts on the network without the requirement that *earlier communications should set up special transmission channels*. Its header format is shown in Figure 4.

One thing that you may notice is that since UDP doesn't require a *formal setting up process* for transmission, it affects the reliability and integrity of the data transmitted. If you wonder why we need that, then I would suggest you think about services that are real-time and those applications in which the waiting for lost packages is not the preferred option.

If you wish to explore more about networking in general, I would suggest you read *Computer Networks* by Andrew Tanenbum.

Linux networking

Let's ponder a little more about networking in the Linux platform. First of all, we will look at some basic commands.

ifconfig: This is used to configure the *kernel-resident network interfaces*. If you wish to display the status of all interfaces (including those that are down), you can issue the following command (please don't skip the results, as that may help you get accustomed to the available interfaces):

```
aasivsinayak@GNU-B0X:~$ ifconfig -a
eth0  Link encap:Ethernet  HWaddr 00:25:64:56:f4:e1
      inet addr:--HIDDEN-- Bcast:131.227.156.255 Mask:255.255.255.0
      inet6 addr:--HIDDEN-- Scope:Link
      UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
      RX packets:371084 errors:0 dropped:0 overruns:0 frame:0
      TX packets:245076 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen: 1000
      RX bytes:476799041 (476.7 MB)  TX bytes:22141513 (22.1 MB)
      Interrupt:29 Base address:0x8000

eth2  Link encap:Ethernet  HWaddr 00:26:5e:7d:8d:53
      inet6 addr: fe80::226:5eff:fe7d:8d53/64 Scope:Link
      UP BROADCAST MULTICAST  MTU:1500  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
```

The Seven Layers of OSI

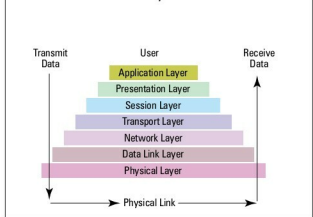


Figure 1: The seven layers of OSI

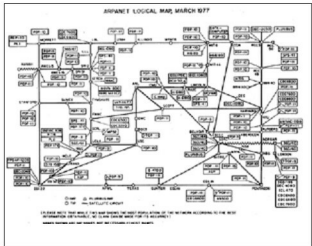


Figure 2: ARPANET logical map, March 1977

(Source: The Computer History Museum)

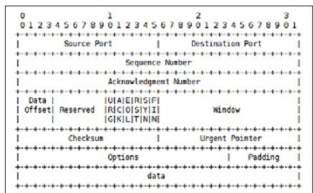


Figure 3: The TCP/IP header format

(Source: RFC791)

```
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen: 1000
RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
Interrupt:17 Base address:0xc000

lo    Link encap:Local Loopback
```